

GAME DESIGN DOCUMENT

Untitled Military Strategy Game

Version 1.2 • June 2026 • Pre-Production

Platform: PC (Steam) • Engine: Godot 4 • Genre: Real-Time Strategy

1. Game Overview

1.1 Concept Summary

A modern military real-time strategy game for PC (Steam) in which 2-8 players load into a procedurally generated map and compete to destroy each other's Headquarters. Players expand their territory by securing Power Crystals, constructing Command Centers, building Powerplants for income, training troops, and researching technologies to gain a strategic edge.

The game is inspired by The Conquerors 3 (Roblox) but targets the Steam PC audience with deeper systems, better balance, and a longer-form strategic experience.

1.2 Core Design Pillars

- Slow burn strategy — matches last 30-60+ minutes, rewarding long-term planning over reaction speed
- Meaningful expansion — Power Crystals (standard and super) are the engine of the entire economy and base-building system. Super Crystals are the highest value objectives on any map — controlling them means faster income, larger bases, and stronger insurance via HQs
- Readable chaos — colorful, arcade-y visuals ensure player armies and structures are always distinguishable at a glance in a multiplayer match
- Accessible depth — soft unit counters mean new players aren't punished for not memorizing matchups, but experienced players can still exploit positioning and composition
- Solo dev simplicity — systems are deliberately kept lean for v1.0. Fog of war, multiple resources, and factions are deferred until the core loop is proven fun
- Per-category unit caps — infantry, tanks, aircraft, and ships each have independent global caps that prevent any single unit type from dominating and create meaningful army composition decisions

1.3 Target Audience

- Steam PC players aged 16-30 who grew up on browser and Roblox strategy games
- Fans of Age of Empires, The Conquerors 3, and Command & Conquer who want a modern accessible multiplayer experience
- Small group friend sessions (2-4 players) as well as public matchmaking

1.4 Platform & Technology

Category	Choice	Reason
Engine	Godot 4	Best-in-class 2D, open source, MIT license, no royalties
Platform	PC (Steam)	Target audience, Steamworks free multiplayer infrastructure

Multiplayer	GodotSteam + Steam Datagram Relay	Free relay servers, no NAT issues, Steam lobby integration
Networking model	Listen server (one player hosts)	No hosting costs, integrates with Steam lobbies
Sync method	Event-driven RPCs + MultiplayerSynchronizer	Efficient for strategy game pace
Movement sync	Client-side prediction + server correction	Responsive feel with authoritative accuracy
Combat	Server authoritative	Prevents desync and cheating
Art tool	Aseprite	Industry standard pixel / sprite art tool
Version control	Git + GitHub	Industry standard, free
Project mgmt	Notion or Trello	Task tracking and milestone planning

2. Core Gameplay

2.1 Win Condition

A player is eliminated when all of their Headquarters (HQ) structures are destroyed. Players start with one HQ but can build additional HQs on any crystal they control, providing insurance against elimination. The last player (or team) with at least one standing HQ wins the match.

Attacking players must locate and destroy every HQ a player owns before they are eliminated. A wealthy player may have 3-4 HQs spread across the map, turning elimination into a sustained campaign. Conversely, a player who skips additional HQs and spends everything on military is making a high-risk high-reward bet.

NOTE: *In team mode, ALL HQs belonging to ALL members of a team must be destroyed to win. A team is not eliminated until every player on that team has lost every HQ.*

2.2 Match Flow

- **Match Start** — Each player spawns on a Super Crystal with a pre-placed HQ and Powerplant. They begin with a Builder and a small infantry squad. Starting money funds early expansion
- **Early Game** — Builders move to nearby crystals. Opening the build menu near a crystal shows placement options highlighted green or red. Priority is securing Powerplants for income and locking down crystals before opponents
- **Mid Game** — Players expand further, build military structures, train armies, and research upgrades. Mid-map Super Crystals become major contested objectives. Players decide whether to invest in additional HQs for insurance or spend on military instead
- **Late Game** — Large armies clash. Players must locate and destroy every enemy HQ to win. Players with multiple HQs force attackers into sustained multi-front campaigns. Naval and air power become decisive

2.3 Player Controls

- Left click — select a single unit or building
- Shift + left click — add or remove individual units from the current selection
- Click and drag — box-select all units within the drawn rectangle
- Right click — issue a move or attack order to selected units
- Camera — WASD or arrow keys to pan; scroll wheel to zoom

NOTE: *Control groups (Ctrl+1-9 to assign, number key to recall) should be implemented to support managing multiple army groups.*

3. Map & World

3.1 Map Generation

Every match generates a new procedural map. Maps scale in size based on the number of players — more players receive a proportionally larger map to ensure adequate expansion space and avoid early forced conflict. Power Crystals and Oil Deposits are distributed evenly across land and water respectively, with spacing algorithms ensuring no player starts with an unfair advantage in nearby crystal access.

NOTE: Exact crystal counts per map size to be determined during playtesting. The goal is that players always have room to expand but must eventually contest crystals with opponents.

3.2 Terrain Types

Terrain	Effect
Open Plains	No movement modifier. Default buildable land.
Forest / Trees	Slows unit movement. May provide light cover. No fog of war in v1.0.
Mountains / High Ground	Defensive advantage for units positioned on them. Impassable or slow to cross.
Rivers / Water	Impassable by ground units. Oil Deposits spawn here. Oil Rigs can be built on water but no Command Centers or land bases.
Roads	Increases unit movement speed when travelling along them.

NOTE: No fog of war in v1.0. All terrain and units are visible to all players at all times. This simplifies the prototype significantly and can be revisited post-launch.

3.3 Power Crystal Types (Land)

Crystal Type	Properties
Standard Crystal	The most common crystal. Provides a default build radius around it. Standard income when a Powerplant is built here. Any player can place a Powerplant on an uncontested standard crystal. Locked to an enemy if they have placed any non-Powerplant structure.

Super Crystal	Rare and high value. Provides a significantly larger build radius than standard crystals. Generates 2-3x the income of a standard crystal when a Powerplant is built here. Guaranteed as each player's starting position. Also appears rarely as mid-map objectives — the most contested locations in any match. Placing an HQ on a Super Crystal provides the largest build radius in the game.
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3.4 Oil Deposits (Water)

Oil Deposits are the water equivalent of Power Crystals, spawning in rivers, lakes, and coastal areas. They follow the same standard / massive size split and income rules. Oil Rigs are the water equivalent of Powerplants. No Command Centers or base structures can be placed on water — Oil Rigs are purely economic outposts and are vulnerable without nearby land-based defenses.

4. Resources & Economy

4.1 Resource Overview

The game uses a single resource: Money. All units, buildings, and research are purchased with money. This keeps the economy simple and readable for players and reduces implementation complexity for v1.0. Additional resources (e.g. oil for vehicles) may be introduced in a future update.

4.2 Income Sources

Source	How It Works
Powerplant (land)	Built on or near a Power Crystal. Generates money passively over time. Can be built without a Command Center nearby — builders can push these out across the map independently.
Oil Rig (water)	Built on an Oil Deposit. Functions identically to a Powerplant but on water. Equally vulnerable without defenses.
Super Crystal income bonus	Powerplants built on a Super Crystal generate 2-3x the income of a standard crystal Powerplant. Oil Deposits do not have a super variant — all water income is equal.
Starting money	Each player begins the match with a small amount of money to construct their first Powerplant and Builder queue.

4.3 Crystal Claiming Rules

- Standard Crystal — any player can place a Powerplant on an uncontested standard crystal. Multiple players can share a contested crystal by both having Powerplants, splitting the income
- Super Crystal — same claiming rules as standard crystals but significantly more valuable. Expect these to be the most fought-over locations on the map
- Locking a crystal — placing any non-Powerplant structure at a crystal prevents opponents from building anything there until all your structures are destroyed
- Contesting — destroying an enemy's Powerplant opens the crystal back up. The crystal itself cannot be destroyed
- Oil Deposits — always contestable since no base structures can be placed on water. Naval control of water areas determines who can safely maintain Oil Rigs

NOTE: The inability to lock down water deposits is intentional. Water income is always an ongoing fight rather than a one-time claim.

4.4 Economic Strategy

The economy creates a natural tension between two approaches:

- **Fast expand** — rush Builders to as many crystals as possible, drop Powerplants everywhere, generate income quickly but leave them undefended and vulnerable to raids
- **Secure expand** — invest in Command Centers to lock down crystals exclusively and build defensive structures, but expand slower and generate income later

Raiding enemy Powerplants is always a viable strategy — destroying income buildings hurts the enemy economy without requiring an assault on their main base.

5. Buildings & Base Construction

5.1 Builder Units

All structures in the game are constructed by Builder units. Builders are trained from the HQ and are the core of all expansion. No building can be placed without a Builder physically constructing it. Builders can be attacked and destroyed, making them a high-value target for disrupting enemy expansion.

5.2 Build Zone System

Most structures can only be placed within the build radius of an HQ or Command Center. The exceptions are:

- Powerplants — can be built anywhere on the map near a Power Crystal, no Command Center required
- Oil Rigs — can be built anywhere on water near an Oil Deposit
- Walls / Fortifications — can be placed anywhere within any friendly build radius

NOTE: This means Builders can be sent deep into the map to secure income without first establishing a full base — but those structures will be exposed and undefended.

5.3 Building Roster

Building	Function	Notes
Headquarters (HQ)	Win condition structure. All of a player's HQs must be destroyed to eliminate them. Trains Builder units. Expands the build radius of the crystal it is placed on.	One HQ pre-placed on starting Super Crystal. Additional HQs built by a Builder on any owned crystal. Expensive by design — building multiple is a meaningful strategic trade-off vs spending on military.
Command Center	Extends build radius into new territory. Locks down a crystal exclusively when placed near one. Has basic defenses.	Must be placed near a Power Crystal. Enables all other structures in its radius. Cannot be placed on water.
Powerplant	Generates money passively. Can be placed near any crystal without a Command Center.	Vulnerable if undefended. Multiple players can share a crystal until a Command Center locks it. 2-3x income near Massive Crystals.
Oil Rig	Water equivalent of a Powerplant. Generates money passively from Oil Deposits.	Water only. Cannot be locked exclusively — always contestable. 2-3x income near Massive Deposits.

Shipyard	Trains all ship units. Built by a Builder on land directly adjacent to water.	No Command Center required. Cannot be placed near enemy structures or units. Ensures naval access regardless of coastal crystal placement. Ships queue here and sail out when complete.
Barracks	Trains infantry units.	Requires Command Center build radius.
Vehicle Depot	Trains ground vehicles (tanks, APCs, jeeps).	Requires Command Center build radius.
Airfield	Trains aircraft (jets, helicopters).	Requires Command Center build radius. High cost.
Research Lab	Unlocks new units, upgrades, and special abilities.	1-2 per Command Center cluster. More bases = faster research.
Wall / Fortification	Defensive barrier. Blocks unit movement.	Has HP and can be destroyed.
Turret	Auto-attacks nearby enemy units.	Stationary defense. Useful for protecting exposed Powerplants and Oil Rigs.

NOTE: All buildings have HP and can be damaged and destroyed. Visible damage states at 50% and 25% HP are recommended.

6. Units & Combat

6.1 Combat Philosophy

Combat uses soft counters — unit type matters but does not override positioning, numbers, and tactical decisions. A player who understands terrain, flanking, and combined arms will beat a player who simply produces more of a counter unit. No unit is useless and no unit is unstoppable.

6.2 Unit Caps

Unit Category	Cap & Notes
Infantry	Starting cap: ~20 units. Trained from Barracks. Includes all infantry types.
Tanks	Starting cap: ~6 units. Trained from Vehicle Depot.
Aircraft	Starting cap: ~4 units. Trained from Airfield. High cost, high impact.
Ships	Starting cap: ~6 units. Trained from Shipyard (name TBD). Water only.
Builders	Separate cap from combat units. Value TBD. Not infantry — trained from HQ.

All caps are global per player across all bases. Caps can be raised beyond starting values — the exact mechanism (research, buildings, or both) is undecided and will be balanced during playtesting.

NOTE: Starting cap values are estimates. Final values to be determined during balance playtesting.

6.3 Infantry Units

Unit	Role	Key Notes
Rifleman	Standard all-purpose infantry. Backbone of the army.	Cheap, trainable in bulk. Effective vs infantry.
Machine Gunner	High damage output, slow movement.	Devastating in defensive positions. Vulnerable when moving.
Sniper	Long range, high single-target damage, very fragile.	Counters high-value infantry. Weak vs vehicles and groups.

RPG Soldier	Anti-vehicle and anti-air infantry.	Soft counter to tanks and low-flying aircraft. Fragile vs infantry.
Engineer	Repairs buildings and vehicles.	Does not construct new buildings — that is the Builder's role. Can repair damaged structures in the field.
Medic	Passively heals friendly units in a radius. No attack order needed.	Stacks with multiple Medics. High value target — enemy will prioritize killing them.
Builder	Constructs all buildings. Core of expansion.	Trained from HQ. Not counted in infantry cap. Multiple Builders speed up construction.

- Infantry can board APCs or Transport Helicopters to move rapidly across the map

6.4 Ground Vehicles

Unit	Role	Key Notes
Jeep / Scout	Fast, light, excellent vision range.	Ideal for early scouting and raiding exposed Powerplants. Weak in direct combat.
APC	Armored Personnel Carrier. Transports up to N infantry.	Infantry inside are protected until they disembark. Can be destroyed with troops inside.
Light Tank	Faster than MBT, less armor. Balanced all-rounder.	Good for flanking and rapid strikes. Countered by RPG Soldiers and Artillery.
Main Battle Tank	Slow, heavy armor, high damage. Frontline powerhouse.	Most durable ground unit. Expensive. Weak to Artillery and coordinated RPG fire.
Anti-Air Vehicle	Counters aircraft. Weak vs ground targets.	Essential if enemy is massing air units. Useless against pure ground armies.
Artillery	Long range, area damage, very fragile.	Must stay behind the front line. Devastating against grouped infantry and static defenses. Countered by fast units like Jeeps.

6.5 Aircraft

Unit	Role	Key Notes
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Fighter Jet	Fast air-to-air combat unit. Hard counters other aircraft.	Fastest unit in the game. Limited effectiveness vs ground. Countered by Anti-Air Vehicles.
Attack Helicopter	Slow, devastating vs ground units and vehicles.	Shreds infantry and light vehicles. Vulnerable to Fighter Jets and Anti-Air.
Bomber	Slow, high area damage, fragile. Targets ground only.	Extremely powerful against grouped units and buildings. Easy target for Fighters and Anti-Air. High priority kill.
Transport Helicopter	Moves infantry rapidly across the map. Unarmed.	Enables surprise flanks and fast reinforcement of distant bases. Destroyed easily if intercepted.

- Aircraft fly indefinitely — no fuel or ammo return mechanic in v1.0
- All aircraft can be attacked by Fighter Jets and Anti-Air Vehicles
- Attack Helicopters and Bombers can engage ground units and buildings
- Carriers (see Ships) act as mobile Airfields and can deploy aircraft from water

6.6 Ships

Unit	Role	Key Notes
Destroyer	Balanced combat ship. Anti-submarine capable.	The workhorse of the naval fleet. Effective vs most ship types.
Battleship	Slow, massive range, heavy damage. Expensive.	Can bombard shore targets and buildings from long range. Devastating but slow to position.
Carrier	Deploys aircraft from water. Acts as a mobile Airfield.	Does not fight directly. Allows air power projection without a land base. Extremely high value target.
Submarine	Stealthy torpedo attacks. Fragile if spotted.	Hidden from enemy vision until it attacks or a Destroyer is nearby. Hard counter to large ships. Countered by Destroyers.
Oil Ship	Constructs Oil Rigs on Oil Deposits. Naval equivalent of Builder.	Unarmed. Must travel to an Oil Deposit and build an Oil Rig to generate income. Vulnerable without escort.

- Ships can attack both naval targets and ground / building targets near the shore
- Submarines are invisible to most units — Destroyers can detect and engage them
- Carriers allow aircraft to be trained and deployed from water — the aircraft still count toward the global aircraft cap

NOTE: Shipyard can be placed by a Builder anywhere on the coast — no Command Center or crystal required. Cannot be placed within a minimum radius of enemy structures or units. This guarantees naval access on any map regardless of coastal crystal distribution.

6.7 Terrain & Combat Interaction

- Mountains — defensive bonus for units on high ground
- Forests — may slow movement; possible light concealment (no fog of war in v1.0)
- Roads — increased movement speed for rapid reinforcement
- Rivers — ground units must use bridges; ships pass freely
- Water — ground units cannot enter water; ships operate exclusively on water

7. Research & Upgrades

7.1 Research Lab Mechanics

The Research Lab provides access to a flat research menu — not a tree. All research options are visible from the start of the match. Players select any available research, pay the money cost, and wait for it to complete over time. There are no prerequisites or unlock paths. Strategy comes from choosing what to prioritize with limited money and research time.

Each Command Center cluster is limited to 1-2 Research Labs. Players with more bases can run more labs simultaneously, researching faster. This is a direct reward for territorial expansion beyond income alone.

NOTE: Research times, costs, and lab limits per cluster to be balanced during playtesting.

7.2 Stackable Researches

Most researches can be purchased multiple times for stacking bonuses. For example purchasing 'Infantry Speed Boost' twice gives double the speed increase. This gives late game players a meaningful money sink and rewards sustained economic dominance.

NOTE: Stack limits and diminishing returns to be determined during balance playtesting.

7.3 Research Categories

Category	What It Covers
Population Cap Increases	Raise the unit cap for each category independently. Infantry cap +5, Tank cap +2, Aircraft cap +1, Ship cap +2 per purchase. Stackable.
Special Unit Unlocks	One-time purchases that unlock elite units not available by default. See Section 7.4 for full list.
Passive Economy Bonuses	Faster Powerplant and Oil Rig income generation. Cheaper unit and building costs. Stackable.
Special Abilities	Unlock powerful one-time or cooldown-based abilities usable from the command interface. See Section 7.5.

7.4 Special Unit Unlocks

Unit	Category	Description
Humvee	Ground Vehicle	Research upgrade to the Jeep. Seats 6 infantry instead of 4, faster movement, and more armor. Replaces the Jeep in the Vehicle Depot once researched.

Special Forces	Infantry	Elite infantry squad. High damage, high HP, and expensive. The most powerful infantry unit. Trained from Barracks after research.
Heavy APC	Ground Vehicle	Larger APC with a mounted gun and increased infantry capacity. Can defend itself unlike the standard APC. Trained from Vehicle Depot after research.
Stealth Bomber	Aircraft	A bomber variant that is significantly harder to detect and intercept. Functions like a standard Bomber but evades Anti-Air and Fighter Jets more effectively. Trained from Airfield after research.
Nuclear Submarine	Naval	A submarine variant that can launch a single devastating missile strike against any target on the map. Extremely expensive. The missile can be intercepted by Anti-Air. After firing, the Nuclear Submarine becomes a standard Submarine. Trained from Shipyard after research.

7.5 Special Abilities

Ability	Type	Description
Airstrike	Cooldown ability	Call in a targeted airstrike on a map location. Deals heavy damage in a small area. Countered by Anti-Air units near the target.
Artillery Barrage	Cooldown ability	Call in a sustained artillery bombardment on a target area. Slower than an airstrike but covers a larger area over several seconds.
Emergency Reinforcements	Cooldown ability	Instantly spawn a small squad of infantry at a chosen friendly Command Center or HQ. Useful for defending a base under sudden attack.

NOTE: Cooldown durations, damage values, and unlock costs to be determined during playtesting. Additional special abilities may be added in future updates.

8. Multiplayer & Match Settings

8.1 Player Count & Map Size

Player Count	Map Size
2 players	Small
3-4 players	Medium
5-6 players	Large
7-8 players	Huge

Map size scales automatically with player count. Crystal count scales proportionally with map size to ensure consistent expansion opportunities regardless of lobby size.

8.2 Match Modes

- Free for All — every player competes independently. Last player with a standing HQ wins
- Team vs Team — players divided into equal teams (2v2, 3v3, 4v4). A team is eliminated only when all HQs belonging to all members are destroyed

NOTE: *Balanced team sizes are enforced in public matchmaking. Uneven team configurations are available in private lobbies at the host's discretion.*

8.3 Lobby System

- Public matchmaking — players queue and are matched with others via Steam matchmaking
- Private lobbies — host creates a lobby and invites friends via Steam. Host has full control over match settings

8.4 Host Controls

Setting	Options
Lobby visibility	Public (open to matchmaking) or Private (invite only via Steam)
Starting money	Host sets the starting money amount for all players. Higher values enable faster early expansion and shorter matches.
Match time limit	Optional. Host can toggle a time limit on or off. If time expires, the player or team controlling the most crystals wins.
Match speed	Normal, Fast, or Slow. Affects all unit movement speeds, build times, research times, and income tick rates simultaneously.

Team lock	Host can lock teams to prevent players from switching sides after assignment.
Unit restrictions	Host can disable specific unit types for the match (e.g. no aircraft, no naval units). Useful for custom game modes.
Kick players	Host can remove players from the lobby before the match begins.

8.5 Disconnection Rules

If a player disconnects mid-match the game continues normally for all remaining players. The disconnected player's units and buildings go neutral — they are no longer commanded by anyone but continue to function normally:

- Neutral units hold their position and auto-attack any unit that enters their range regardless of faction
- Neutral buildings continue generating income (Powerplants) and auto-attacking (Turrets) but cannot train new units or begin new research
- Neutral assets cannot be captured or taken over by any player
- The disconnected player's HQs must still be destroyed to eliminate them — their neutralized empire remains a real obstacle on the map

NOTE: This prevents players from benefiting from an opponent's disconnect while also ensuring rage-quitting doesn't create a free base for enemies to inherit.

8.6 Disconnection Income Sharing (Team Mode)

In team mode, when a player disconnects their Powerplant income is shared with surviving teammates rather than being lost entirely. This keeps disconnects from creating unwinnable economic disadvantages for the remaining team.

- Income from the disconnected player's surviving Powerplants and Oil Rigs is split evenly between all surviving teammates
- If an enemy destroys a disconnected player's Powerplant or Oil Rig, that structure's income share is permanently lost — it no longer pays out to anyone
- Surviving structures continue paying out until destroyed, giving enemies a clear strategic incentive to raid the ghost economy
- The disconnected player's HQs still must be destroyed for elimination — the income share continues as long as their Powerplants survive, even after their HQs fall

Example: Player C disconnects in a 3v3 match with 4 Powerplants generating 100 money/tick each. Players A and B each receive 200 extra money/tick split from C's 400 total. Enemies destroy 2 of C's Powerplants — now A and B only receive 100 extra money/tick split between them from C's 2 remaining structures.

NOTE: Income sharing only applies in team mode. In FFA, a disconnected player's income benefits no one.

8.7 Technical Multiplayer Stack

System	Implementation
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Network topology	Listen server — one player hosts, others connect
Relay	Steam Datagram Relay via GodotSteam (free, no NAT issues)
Lobby system	Steam Lobbies via GodotSteam
Sync method	Event-driven RPCs + Godot MultiplayerSynchronizer
Movement	Client-side prediction with server correction
Combat resolution	Server authoritative — host resolves all combat outcomes
Bandwidth optimisation	Interest management — clients only receive relevant data
Prototype networking	Godot ENet over LAN / direct IP for early testing

9. UI, Art & Audio

9.1 Visual Perspective

Isometric 2D — angled overhead perspective in the style of Age of Empires 2. Gives the game visual depth and personality while remaining manageable for a solo developer.

9.2 Art Style

Colorful and arcade-y. Each player is assigned a distinct team color that tints their units and structures. Power Crystals should be visually distinct and prominent on the map — players should be able to spot them from a distance and immediately understand their value. Massive Crystals are visually larger and more impressive than standard ones.

9.3 HUD Elements

- Money counter — current money total displayed prominently at the top of the screen
- Unit cap counters — displays current vs. maximum for each category: Infantry, Tanks, Aircraft, Ships
- Minimap — bottom corner, full map visible (no fog of war). Shows all units and structures color-coded by player
- Selection panel — appears at bottom when units or buildings are selected
- Alert system — notifications for attacks on friendly structures, completed research, finished training

9.4 Audio Direction

- Music — upbeat strategic soundtrack in the style of Age of Empires. Energetic but not distracting
- UI sounds — crisp click and confirmation feedback for all menu interactions
- Unit acknowledgement — units respond audibly when selected and ordered
- Combat sounds — explosions, gunfire, and vehicle audio tied to on-screen action
- Construction sounds — satisfying building placement and completion audio

NOTE: Voice lines for units are out of scope for v1.0.

10. Counter System

10.1 Philosophy

Combat uses soft counters — unit type creates advantages and disadvantages but never guarantees an outcome. Positioning, numbers, terrain, and Medic support all influence the result. A player who understands combined arms and flanking will consistently beat a player who relies purely on counter units.

The counter relationships below represent design intent. Exact stat values will be tuned during playtesting.

10.2 Counter Relationships

Unit	Strong Against	Weak Against
Rifleman	Infantry	Vehicles, Aircraft
Machine Gunner	Grouped infantry	Vehicles, Aircraft
Sniper	Single high-value infantry targets	Groups, Vehicles
RPG Soldier	Tanks, Aircraft	Infantry groups
Engineer	— (repairs only)	Everything
Medic	— (heals only)	Everything
Special Forces	Infantry, Light vehicles	Heavy tanks, Aircraft
Jeep / Humvee	Exposed infantry, Artillery	Tanks, RPG Soldiers
APC / Heavy APC	Infantry transport	Tanks, RPG Soldiers
Light Tank	Infantry, Jeeps	MBT, Artillery, RPGs
Main Battle Tank	Most ground units	Artillery, RPG groups, Aircraft
Anti-Air Vehicle	All aircraft	Tanks, Infantry
Artillery	Grouped units, Buildings	Jeeps, Aircraft
Fighter Jet	All aircraft	Anti-Air Vehicles
Attack Helicopter	Infantry, Light vehicles	Fighter Jets, Anti-Air
Bomber	Buildings, Grouped units	Fighter Jets, Anti-Air
Stealth Bomber	Buildings, Grouped units	Fighter Jets (reduced effectiveness)
Transport Helicopter	— (transport only)	Fighter Jets, Anti-Air

Destroyer	Most ships, Submarines	Battleships
Battleship	Shore targets, Buildings, Ships	Submarines, Aircraft
Carrier	— (air deployment only)	Submarines, Aircraft
Submarine	Large ships, Carriers	Destroyers
Nuclear Submarine	Any single target (missile)	Destroyers, Anti-Air (missile intercept)
Oil Ship	— (economic only)	Everything

NOTE: These relationships are design intent only. All values subject to change during playtesting and balance passes.

11. Edge Cases & Win Condition Rules

11.1 Simultaneous HQ Destruction

If two players destroy each other's last HQ at the exact same moment the match declares a draw for both players. In team mode, if the last HQ of both teams is destroyed simultaneously the entire match is declared a draw. Both sides are shown on the results screen as tied.

11.2 Host Migration

If the host player is eliminated or disconnects, a new host is automatically assigned from the remaining connected players. The match continues without interruption. GodotSteam's Steam Datagram Relay handles the underlying connection rerouting.

***NOTE:** Host migration is a known implementation challenge in Godot. The listen server model means the host runs authoritative game logic — migrating this cleanly requires saving and transferring game state. This should be prototyped and tested early.*

11.3 Surrender Rules

- **Solo player surrender** — any solo player can surrender at any time from the menu. All of their units and buildings are immediately removed from the map. They are fully eliminated and the match continues
- **Team surrender** — a surrender vote is initiated by one player and requires all surviving teammates to agree. If all agree, the entire team surrenders simultaneously and all their assets are removed. If any teammate declines, the vote fails
- **No forced surrender** — players cannot be voted out of a match by opponents. Surrender is always voluntary

11.4 Last Player Standing Edge Cases

- If all remaining players are on the same team (all opponents eliminated), the match ends immediately and that team wins — even if they haven't destroyed every neutral or disconnected player's HQ
- If a time limit expires and multiple players or teams are tied on crystal count, the tiebreaker method is TBD — candidates include total structures owned, total units alive, or a sudden death period
- A player with no HQs remaining but still having units and buildings on the map is fully eliminated — their remaining assets go neutral immediately upon their last HQ being destroyed

12. Development Roadmap

12.1 Phase 1 — Pre-Production (Current)

- Finalize Game Design Document
- Define full unit roster and tech tree
- Set up Godot 4 project with Git version control
- Establish art style with placeholder sprites

12.2 Phase 2 — Prototype

- Basic isometric map with terrain tiles and Power Crystals
- Builder unit: select, move, construct a Powerplant
- Powerplant generates money over time
- One combat unit: select, move, attack
- HQ placed at start, destruction ends the game
- Local multiplayer over LAN (ENet)
- Answer: is the core loop fun?

12.3 Phase 3 — Alpha

- Full building roster implemented
- Command Center build zone system
- Oil Deposits and Oil Rigs on water
- Massive Crystal / Massive Deposit income bonus
- Procedural map generation with balanced crystal placement
- Medic and Builder as specialized units
- Shipyard placed on coastal land within Command Center radius, ships sail out on completion
- Oil Ship unit and Oil Rig construction on water
- Submarine stealth and Destroyer detection system
- Per-category unit cap system with UI counters
- Research Lab UI and flat research menu
- At least one special unit unlock implemented (Humvee recommended for simplicity)
- GodotSteam integration and Steam lobbies

12.4 Phase 4 — Beta

- Full unit roster and research tree
- Art pass — replace placeholders with final sprites
- Audio implementation
- Balance pass based on playtesting
- Steam Playtest opened to external players

12.5 Phase 5 — Launch

- Bug fixing and performance optimisation
- Steam page with screenshots and trailer
- \$100 Steam Direct fee paid
- Release

12.6 Estimated Timeline

Phase	Estimated Duration
Pre-Production	1-2 months
Prototype	1-2 months
Alpha	4-6 months
Beta	4-6 months
Launch Prep	2-3 months
Total	12-19 months (solo developer)

13. Open Questions & Deferred Decisions

The following items are intentionally deferred — they depend on prototyping results or are lower priority than core systems.

- Full unit roster — infantry types, vehicle lineup, aircraft types, stats, and costs
- Full tech tree — research paths, unlock order, costs, and research times
- Power Crystal counts per map size — determined during playtesting
- Starting money default value — the baseline amount before host adjusts TBD during playtesting
- Match speed multipliers — exact speed values for Fast and Slow modes TBD
- Time limit tiebreaker — if players are tied on crystal count when time expires, resolution method TBD
- Disconnection income share timing — does the share kick in immediately on disconnect or after a short grace period to handle accidental drops TBD
- Super Crystal income multiplier — 2-3x is the target range, exact value needs balancing during playtesting
- Population cap expansion — which buildings increase the cap and by how much
- Crystal build radius values — exact tile radius for standard, super, and HQ-expanded crystals TBD in engine
- HQ cost — expensive enough that multiples are a real sacrifice but not so costly it is never worth building extras TBD
- Super crystal mid-map count — how many super crystals appear outside starting positions per map size TBD
- DEFERRED — Naval system (Shipyards, all ship units, Oil Ships, Oil Rigs) is fully designed but deferred to a post-launch update. All naval content in this document serves as the future implementation spec.
- KNOWN RISK — Snowball / runaway leader problem. A player who gains an early crystal advantage may compound it into an unwinnable situation for opponents. Anti-snowball mechanic to be designed after first playtest sessions reveal the true severity of the problem.
- Research stack limits and diminishing returns — how many times can each stackable research be purchased TBD
- Nuclear Submarine missile interception mechanic — exact Anti-Air interaction TBD
- Stealth Bomber detection threshold — how much harder is it to detect vs standard Bomber TBD
- APC infantry capacity — how many infantry units can board a single APC TBD
- Carrier aircraft limit — how many aircraft a Carrier can deploy simultaneously TBD
- Submarine detection range for Destroyers TBD
- Builder unit cap — separate limit from combat units, value TBD
- Fog of war — removed for v1.0 simplicity; may be added in a future update
- Multiple resources — oil for vehicles deferred; single money resource for v1.0
- Unit voice lines — out of scope for launch, possible post-launch addition
- AI opponents — no solo/offline play currently designed. A basic AI opponent would significantly improve the experience when matchmaking queues are thin, especially at launch. Deferred but strongly recommended for a post-launch update.
- Console ports — not planned; possible long-term if game succeeds on Steam
- Monetisation — free on Steam; cosmetic DLC or battle pass possible post-launch